Systemadministration

Introduktion

Servrar och filsystem

Agenda

Redundant lagring och filsystem över ett nätverk

1. Client-Server
2. NAS
3. SAN
4. RAID
5. RAID with Parity
7. Hardware & Software RAID
6. SMB

1. Client-Server Model

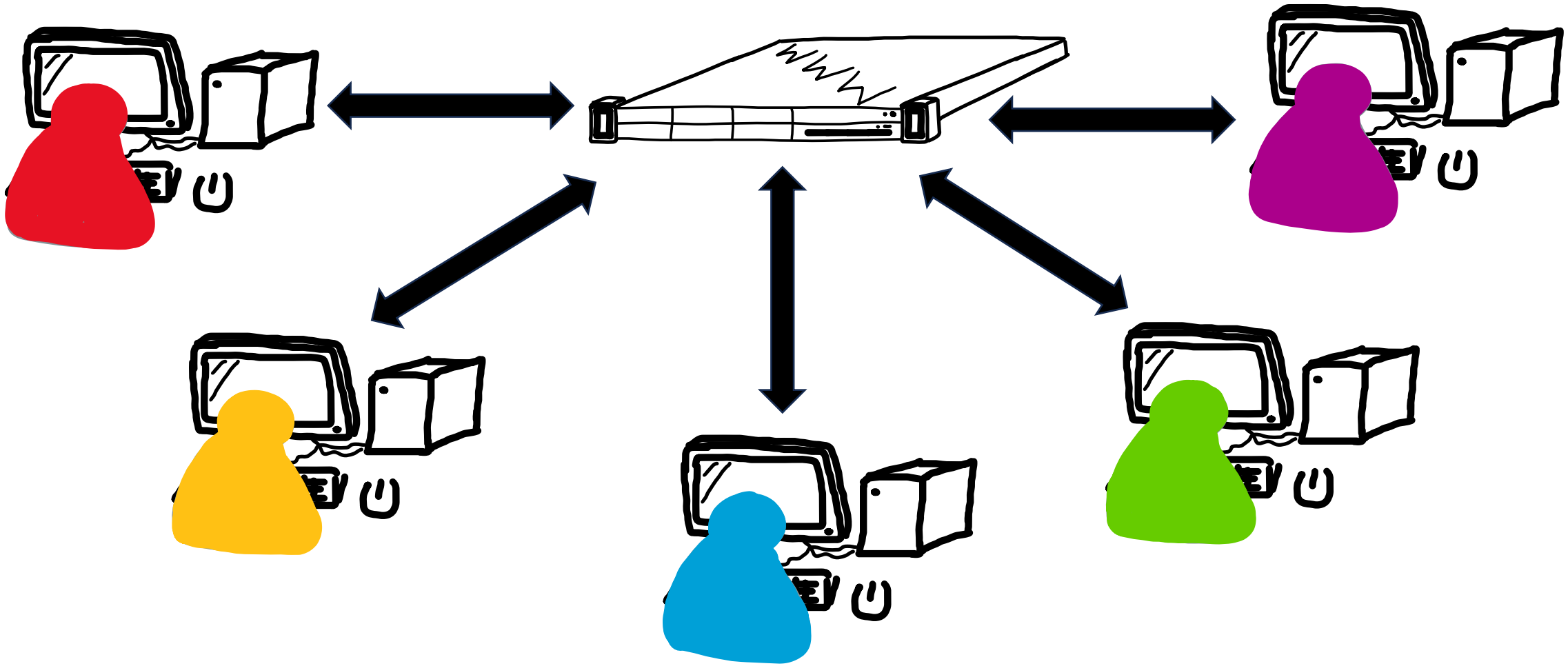
Server

Server

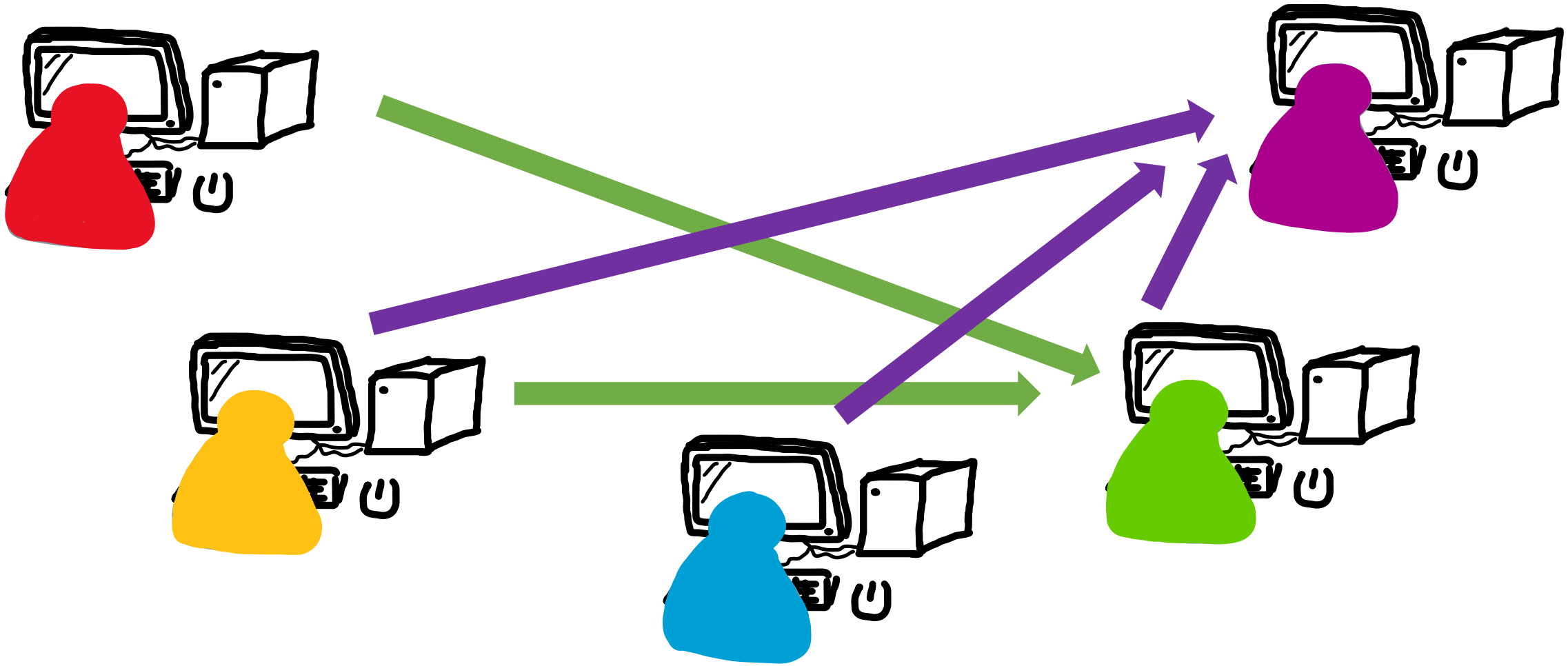
En dator eller mjukvara (bakgrundsprocess) som ’serverar’ en tjänst eller en resurs åt en **klient**

T.ex. filserver, webbserver…

Client-Server Model



Peer-to-Peer (P2P) Model



2. Network Attached Storage

Network-Attached Storage

My first NAS..., by Andrew Curry, 2010, Flickr (http://flic.kr/p/7VqsLZ).CC BY-SA 2.0.

Lagringsserver
delar filer över ett nätverk

Främst för mindre företag & hobby-användare

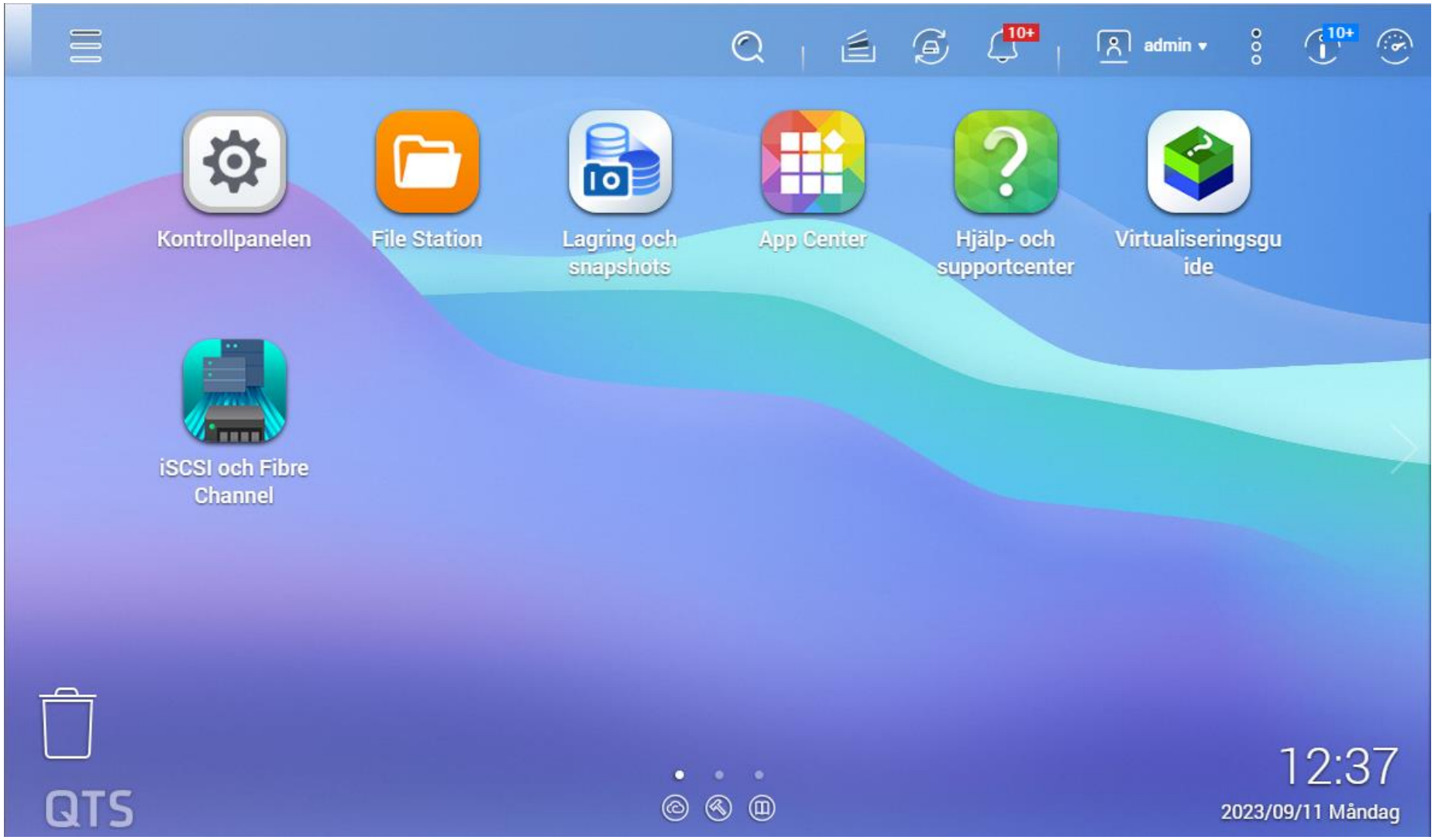
Networked File-Level Storage

Typiskt sett ger en NAS ut **file-level storage**

En del av serverns *filsystem* är tillgängligt över nätverket

Protocol	Native Operating System
SMB	Microsoft Windows
CIFS	Legacy Microsoft Windows
NFS	UNIX/BSD/Linux
AFP	Apple macOS

Network-Attached Storage



3. Storage Area Network

Storage Area Network

Typiskt ett lagringsnätverk med SAN-servrar

Dyrt, komplext, snabbt, mycket lagring & byggt för redundans

Networked Block-Level Storage

Typiskt sett ger ett SAN ut **block-level storage**

En del av serverns *diskyta* görs tillgänglig över ett nätverk

Protocol
Fibre Channel
iSCSI
Infiniband

Obs! Inte exakt samma **block** som i filesystem (men fungerar på liknande vis), utan mer likt **sector**

4. RAID

HDD & SSD Reliability

Exos X20

Capacity

20 TB, 18 TB

Performance, Sustained Data Rate (SDR)

Up to 285 MB/s

MTBF

2.5M hours

Limited Warranty

5 years

Exos X Series, by Seagate, u.å., Seagate (https://www.seagate.com/gb/en/products/enterprise-drives/exos-x/). Fair use/ I enlighet med citaträtt.

Nytro 3050 Series

Capacity

Up to 15.36 TB

Endurance

Up to 10 DWPD

Sequential Read

Up to 2,200 MB/s

Warranty

5 years

Nytro SAS SSD Series, by Seagate, u.å., Seagate (https://www.seagate.com/gb/en/products/enterprise-drives/nytro-sas-ssd/). Fair use/ I enlighet med citaträtt.

SSD Reliability

Backblaze Annual SSD Failure Rates for 2020, 2021, and 2022

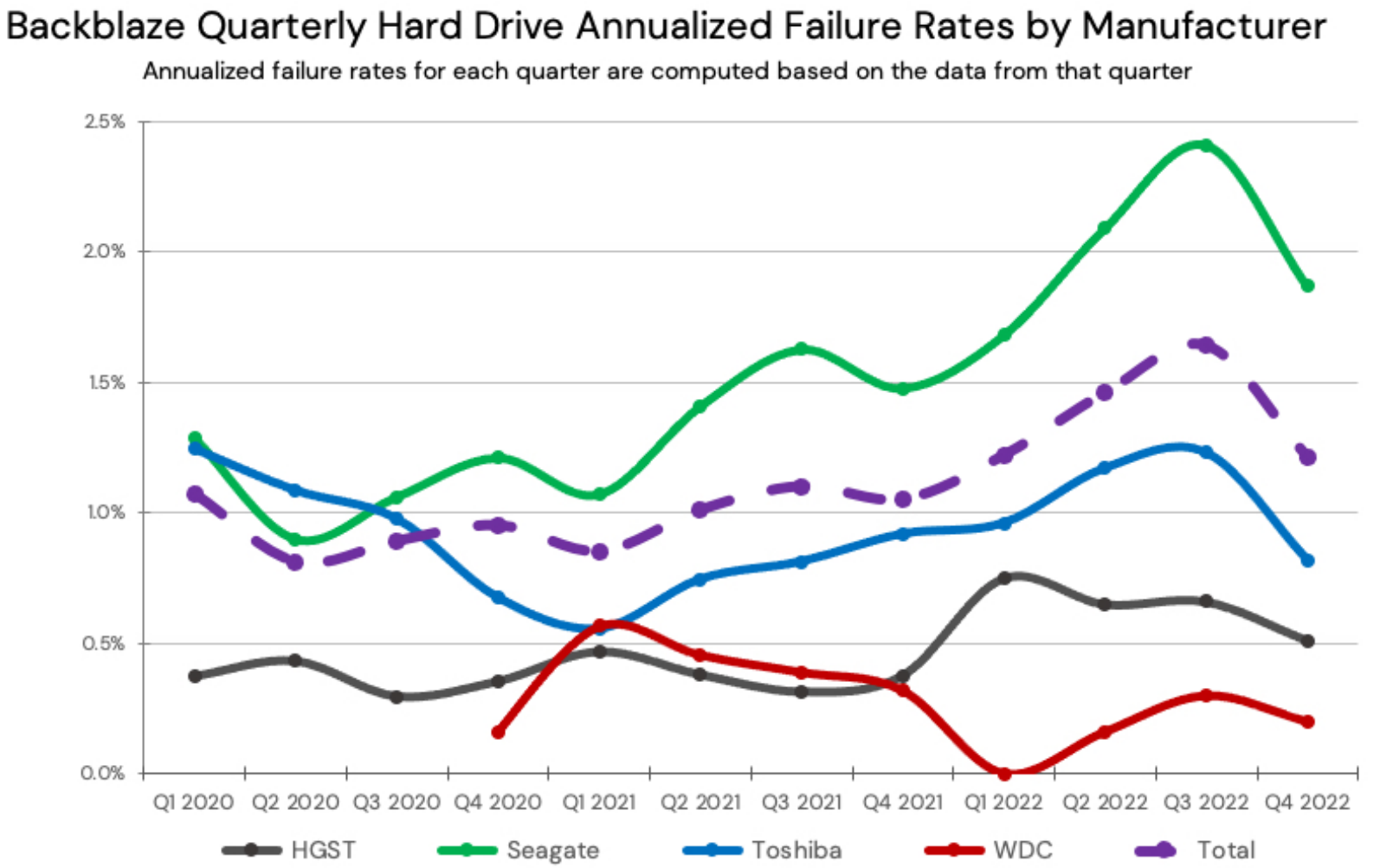
Reporting periods are for each calendar year inclusive

			Annual 2020			Annual 2021			Annual 2022		
MFG	Model	Size (GB)	Drive Days	Drive Failures	AFR	Drive Days	Drive Failures	AFR	Drive Days	Drive Failures	AFR
Crucial	CT250MX500SSD1	250				1,689	2	43.22%	105,365	3	1.04%
Dell	DELLBOSS VD	500	31,566	0	0.00%	63,710	0	0.00%	123,370	0	0.00%
Micron	MTFDDAV240TCB	240	1,473	0	0.00%	33,478	7	7.63%	32,471	1	1.12%
Seagate	SSDSCKKB480G8R	500	4	0	0.00%	256	0	0.00%			
Seagate	ZA250CM10003	250	47,161	0	0.00%	276,281	5	0.66%	400,246	8	0.73%
Seagate	ZA500CM10003	500							591	0	0.00%
Seagate	ZA2000CM10002	2,000	1,460	0	0.00%	1,267	1	28.81%	1,091	0	0.00%
Seagate	ZA250CM10002	250	154,144	2	0.47%	204,287	2	0.36%	203,024	11	1.98%
Seagate	ZA500CM10002	500	6,583	0	0.00%	6,515	0	0.00%	6,544	0	0.00%
Seagate	ZA500GM10001	500							320	0	0.00%
Seagate	ZA250NM1002	250							4,090	0	0.00%
Seagate	SSD	250	39,407	1	0.93%	39,147	1	0.93%	38,527	1	0.95%
WDC	WD Blue SA510 2.5	250							102	0	0.00%
WDC	WDS250G2B0A	250				1,469	0	0.00%	18,700	1	1.95%
		Totals	**281,798**	**3**	**0.39%**	**627,843**	**18**	**1.05%**	**934,441**	**25**	**0.98%**

Backblaze Annual SSD Failure Rates for 2020, 2021, and 2022, by Andy Klein, 2023, Backblaze (https://www.backblaze.com/blog/ssd-edition-2022-drive-stats-review/). Fair use/ I enlighet med citaträtt.

Backblaze

HDD Reliability



	Q1 2020	Q2 2020	Q3 2020	Q4 2020	Q1 2021	Q2 2021	Q3 2021	Q4 2021	Q1 2022	Q2 2022	Q3 2022	Q4 2022
HGST	0.37%	0.43%	0.30%	0.35%	0.47%	0.38%	0.31%	0.37%	0.75%	0.65%	0.66%	0.51%
Seagate	1.29%	0.90%	1.06%	1.21%	1.07%	1.41%	1.63%	1.48%	1.68%	2.09%	2.41%	1.87%
Toshiba	1.24%	1.09%	0.98%	0.67%	0.56%	0.75%	0.81%	0.92%	0.96%	1.17%	1.23%	0.82%
WDC				0.16%	0.57%	0.46%	0.39%	0.32%	0.00%	0.16%	0.30%	0.20%
Total	**1.07%**	**0.81%**	**0.89%**	**0.95%**	**0.85%**	**1.01%**	**1.10%**	**1.05%**	**1.22%**	**1.46%**	**1.64%**	**1.21%**

Backblaze Quarterly Hard Drive Annualized Failure Rates by Manufacturer, by Andy Klein, 2023, Backblaze (https://www.backblaze.com/blog/backblaze-drive-stats-for-2022/). Fair use/ I enlighet med citaträtt.

Backblaze

Redundant **A**rray of **I**ndependent **D**isks

RAID används för

- redundans
- bättre **I/O**-prestanda

Kombinerar flera diskar till en logisk

- **RAID Array** eller **RAID Volume**

RAID Standard Levels

Level	Data Distribution
RAID 0	Striping
RAID 1	Mirroring
RAID 10	Striping + Mirroring
RAID 5	Striping with distributed parity
RAID 6	Striping with double distributed parity

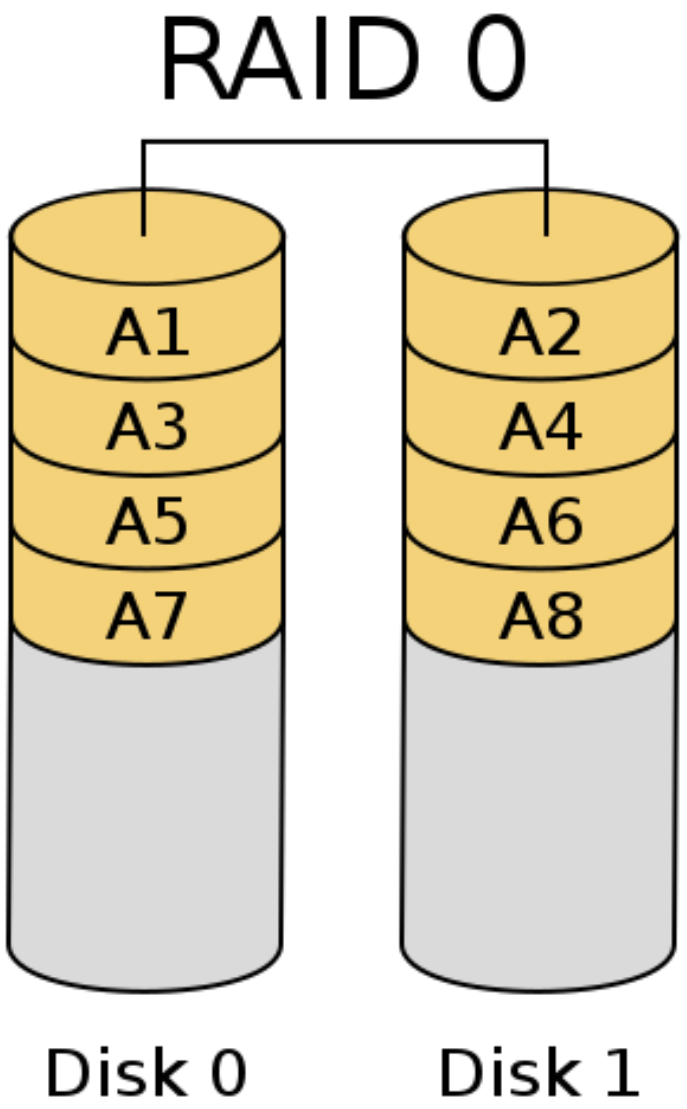
RAID 0

Striped array

Data lagras utspritt på alla diskar

Borde man använda **RAID 1**?

NEJ!
Vid minsta fel förlorar du allt



RAID 0 with two disks (disk 0 and disk 1) over one logical volume A with odd blocks on disk 0 and even blocks on disk 1, by Cburnett, 2006, Wikimedia Common (https://commons.wikimedia.org/w/index.php?curid=1509075). CC BY-SA 3.0.

RAID 0

Minimum number of drives	2 drives
Fault tolerance	0 drives
Read performance (simplified)	$num.drives \cdot speed$, i.e. nx
Write performance (simplified)	$num.drives \cdot speed$, i.e. nx

Ex.

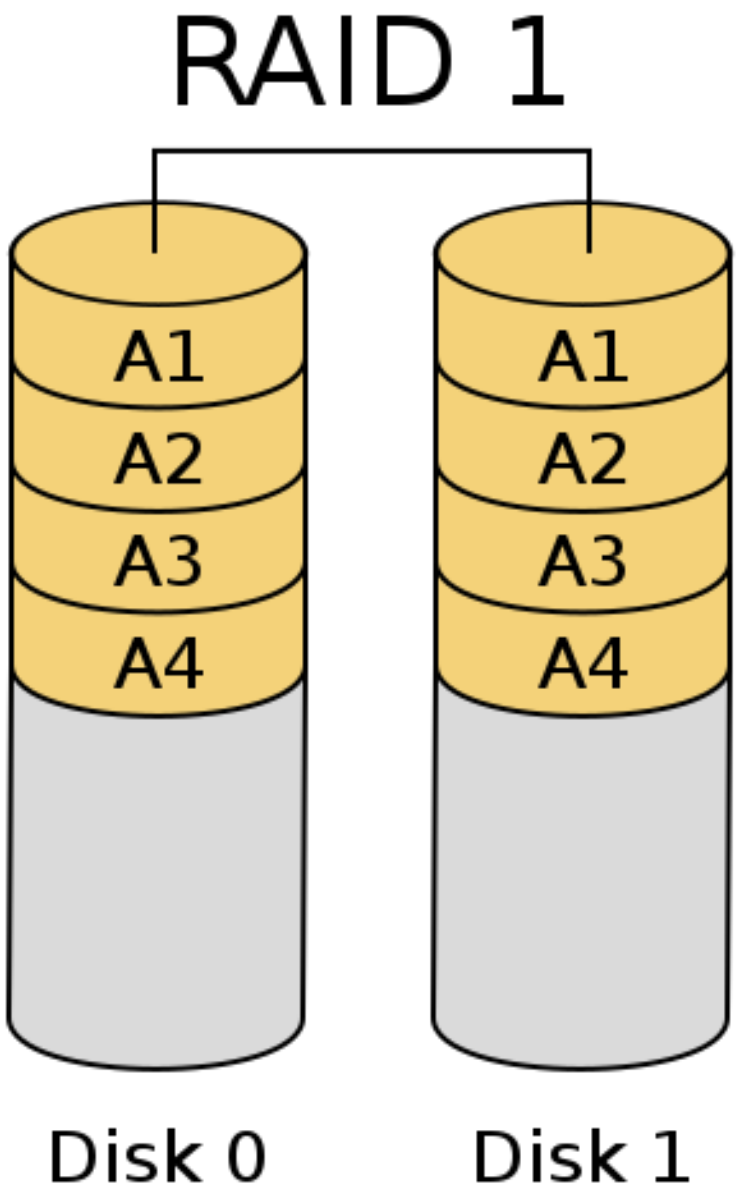
2 diskar, max skrivhastighet 220MB/s

RAID 0: $2 \cdot 220 = 440\ MB/s$

RAID 1

Mirrored array

Samma data lagras på flera diskar, 1:1



RAID 1 with two disks (disk 0 and disk 1) over one logical volume A with all blocks replicated/mirrored from drive 0 to drive 1, by Cburnett, 2006, Wikimedia Common (https://commons.wikimedia.org/w/index.php?curid=1509082). CC BY-SA 3.0.

RAID 1

Minimum number of drives	2 drives
Fault tolerance	$number\ of\ drives - 1$
Read performance (simplified)	$slowest\ drive$, i.e. x
Write performance (simplified)	$num.drives \cdot speed$, i.e. nx

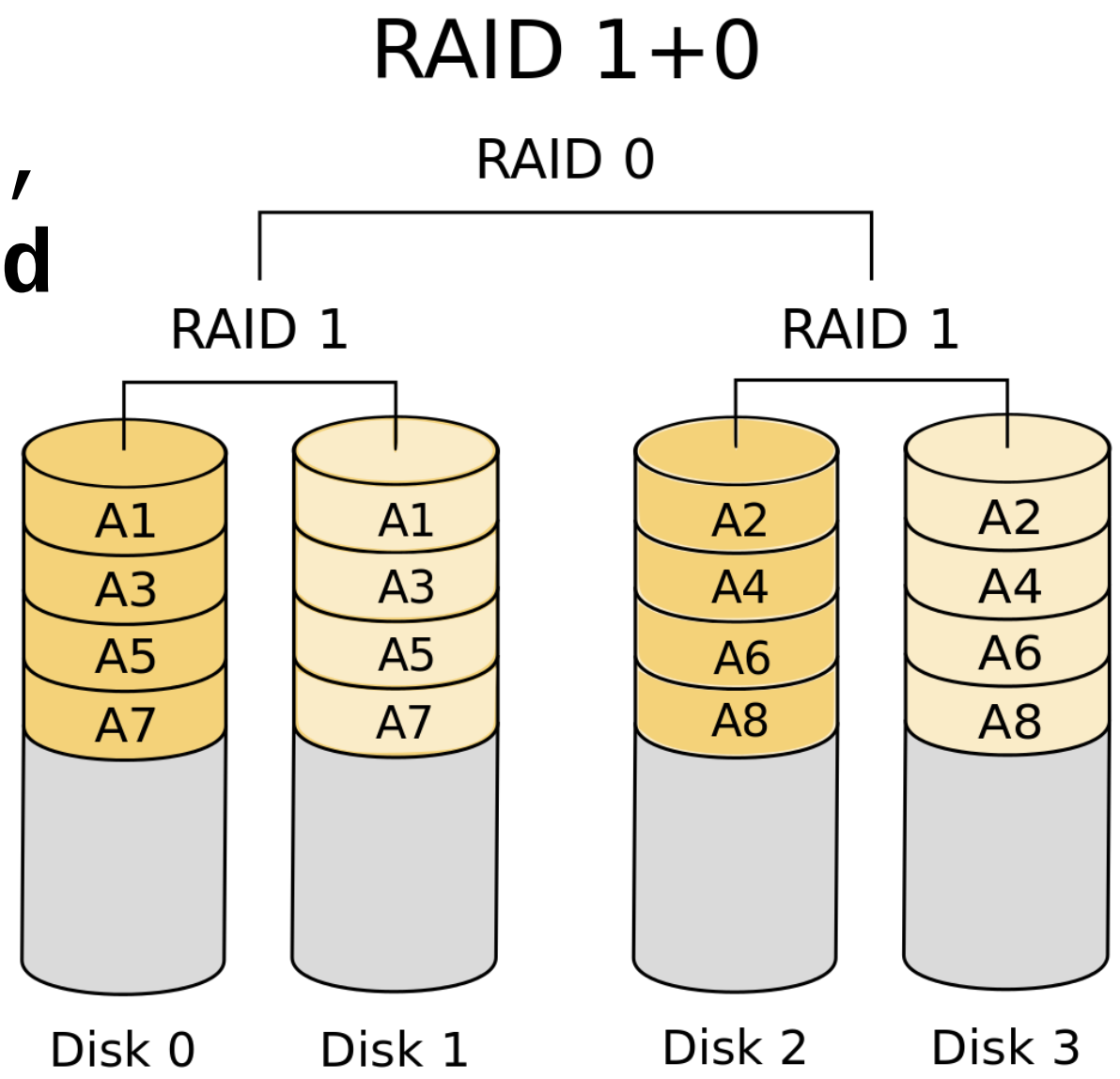
Ex.

3 diskar, hur många kan fallera?

RAID 1: $3 - 1 = 2\ drives$

RAID 10

Två **mirrored arrays**,
som sedan är **striped**



RAID 10 Konfiguration, by Cburnett, 2012, Wikimedia Common (https://commons.wikimedia.org/w/index.php?curid=22591138). CC BY 3.0.

5. RAID with Parity

Parity

Paritet

Ex. Om jag vill spara 101010

Och lägger till paritet kan enkla fel upptäckas

Paritet(101010): 3x 1 = 3 → udda

1	0	1	0	1	0	1

Ett tal med paritet ska alltid vara jämt

Parity

Paritet

1	0	1	0	1	0	1

1	0	1	0	?	0	1

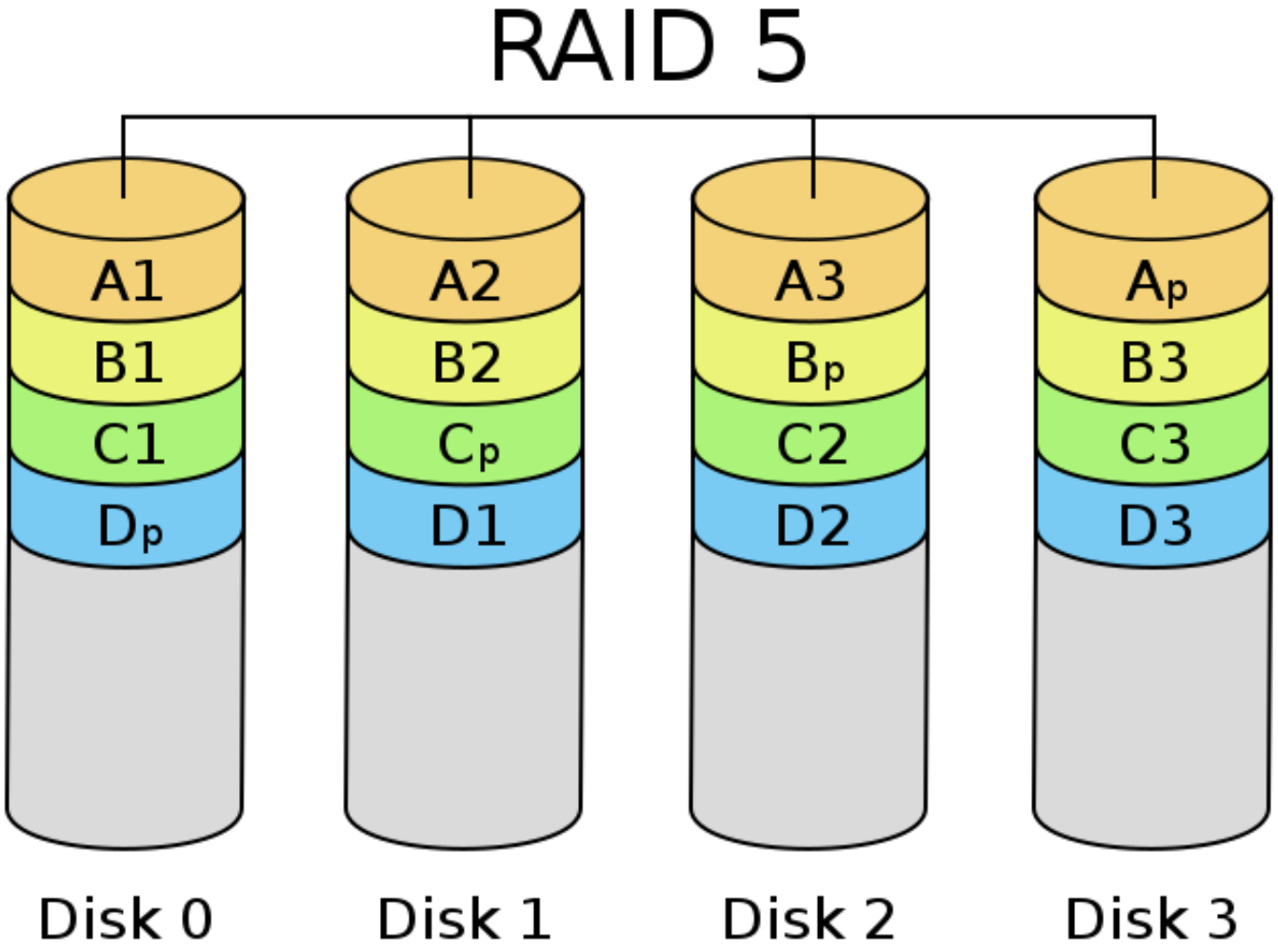
1010?01 → 3x 1 = 3 → udda

? måste vara 1!

RAID 5

Striped array with distributed parity

Om en disk går förlorad kan datan återskapas



RAID 5 with these four disks (disk 0, 1, 2, and 3) and each group of blocks (orange, yellow, green, and blue) have a distributed parity block that is distributed across the four disks., by Cburnett, 2006, Wikimedia Common (https://commons.wikimedia.org/w/index.php?curid=1509158). CC BY-SA 3.0.

RAID 5

Minimum number of drives	3 drives
Fault tolerance	1 drives
Read performance (simplified)	$num.drives \cdot speed$, i.e. nx
Write performance (simplified)	$num.drives - 1 \cdot speed$, i.e. $(n-1)x$

Ex.

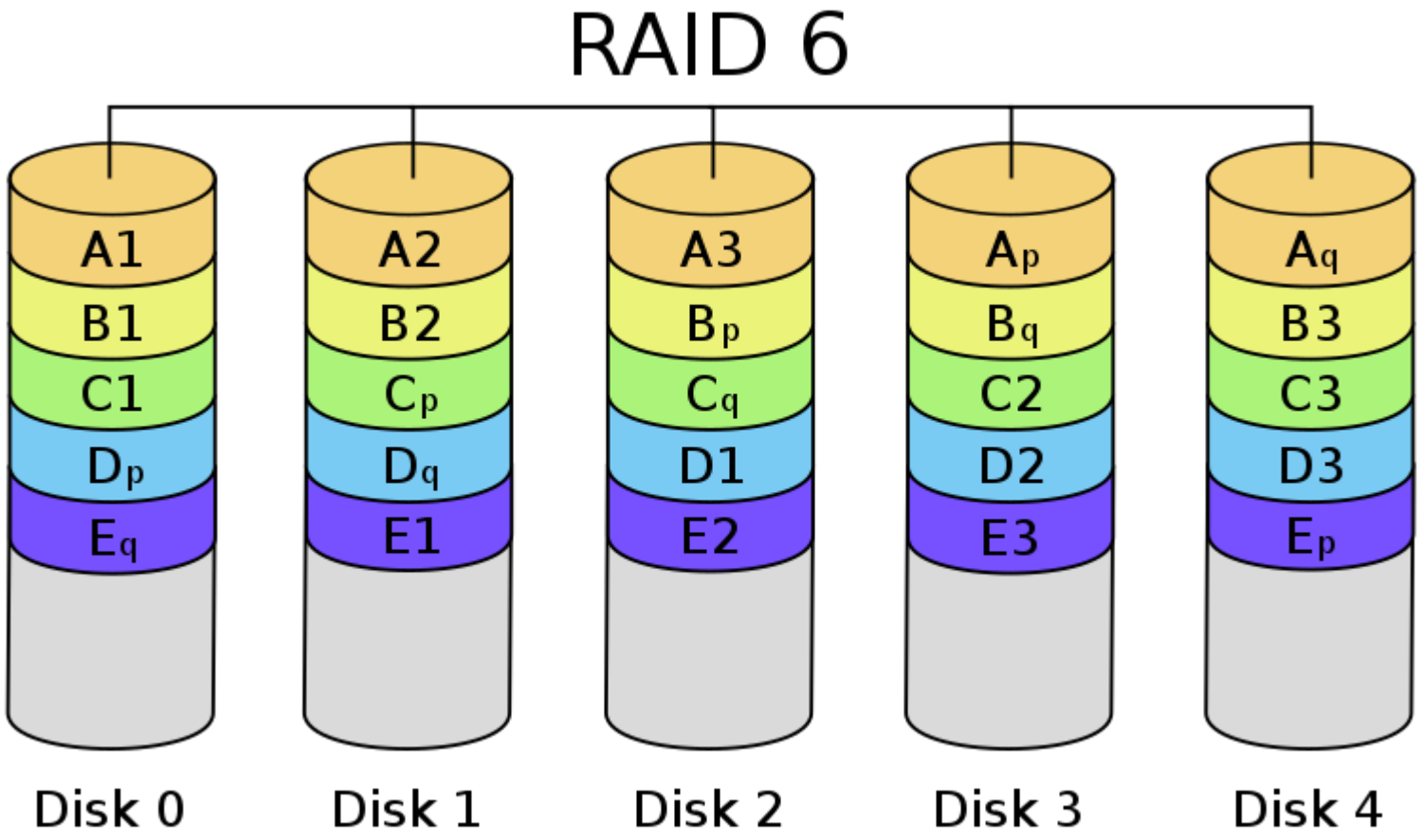
4 diskar, max skrivhastighet 210MB/s

RAID 5: $(4-1) \cdot 210 = 630\ MB/s$

RAID 6

Striped array with double distributed parity

Om två diskar går förlorad kan datan återskapas



RAID 6 with five disks (disk 0, 1, 2, 3, and 4) and each group of blocks (orange, yellow, green, blue, and purple) has two distributed parity blocks that are distributed across the five disks., by Cburnett, 2006, Wikimedia Common (https://commons.wikimedia.org/w/index.php?curid=1509166). CC BY-SA 3.0.

RAID 6

Minimum number of drives	3 drives
Fault tolerance	2 drives
Read performance (simplified)	$num.drives \cdot speed$, i.e. nx
Write performance (simplified)	$num.drives - 2 \cdot speed$, i.e. $(n-2)x$

Ex.

4 diskar, max skrivhastighet 210MB/s

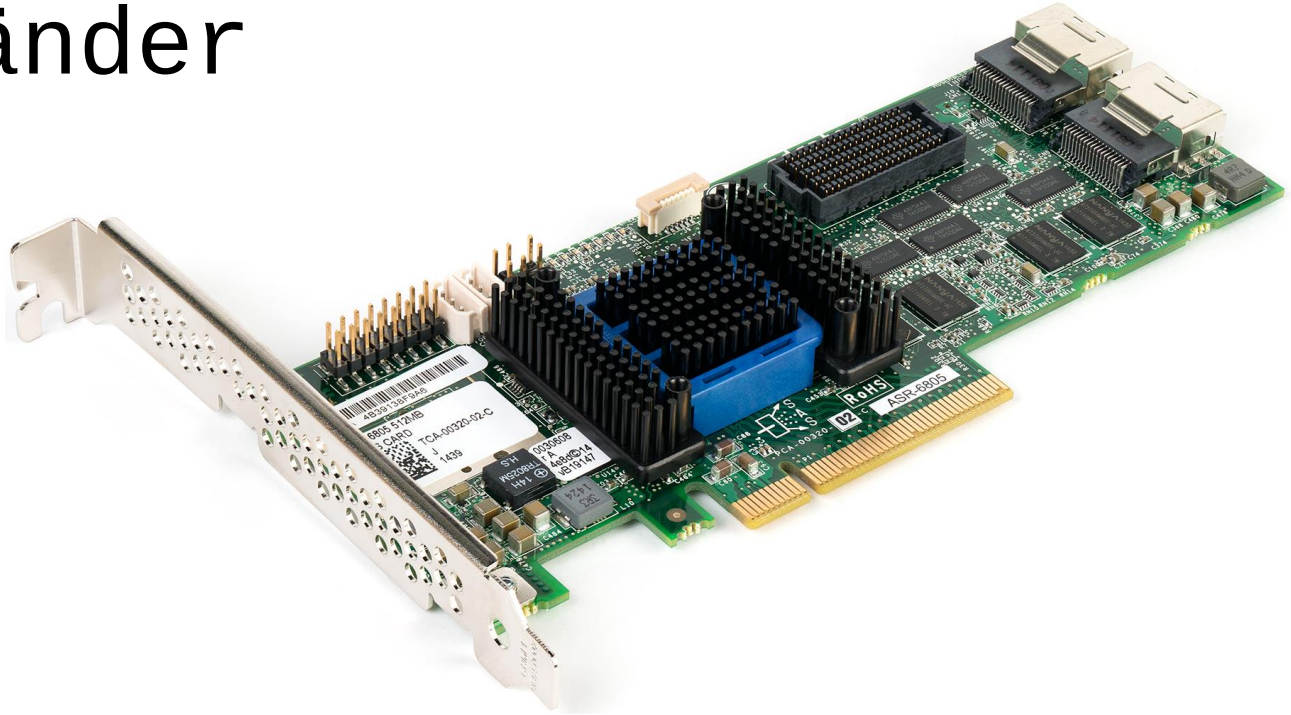
RAID 6: $(4-2) \cdot 210 = 420\ MB/s$

6. HW & SW RAID

RAID Controller

Hårdvaru-RAID använder sig utav en RAID-controller

- Expansionskort
- Integrerad i moderkort

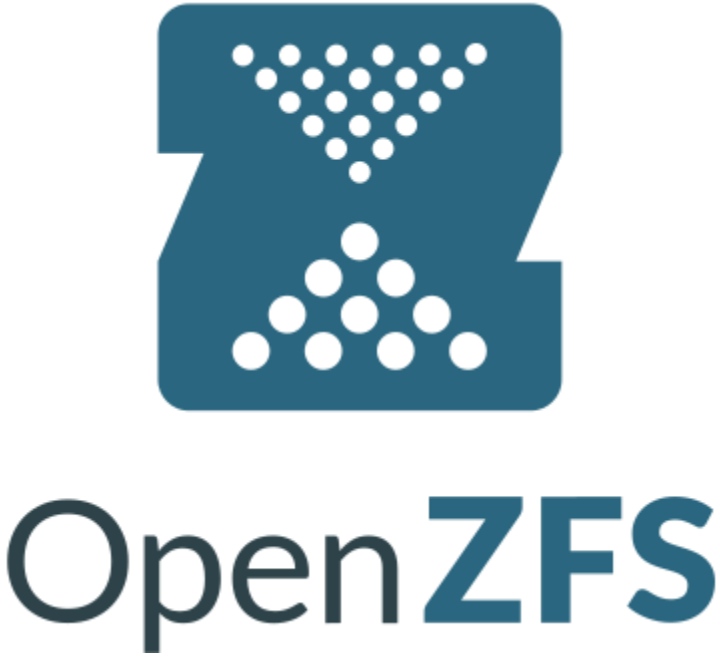


*Adaptec 6805 SAS RAID controller. PCI Express 2.0 x8 expansion card. * Two x4 mini-SAS internal connectors (SFF-8087) *512MB cache * RAID 0, 1, 1E, 5, 5EE, 6, 10, 50, 60 and JBOD* , by Dmitry Nosachev, 2016, Wikimedia Common (https://commons.wikimedia.org/w/index.php?curid=47578838). CC BY-SA 4.0.

Software RAID

RAID hanteras i mjukvara och inte i dedikerad hårdvara

- Mer prestandakrävande
- Mer flexibelt, ofta mer än standard **RAID-levels**



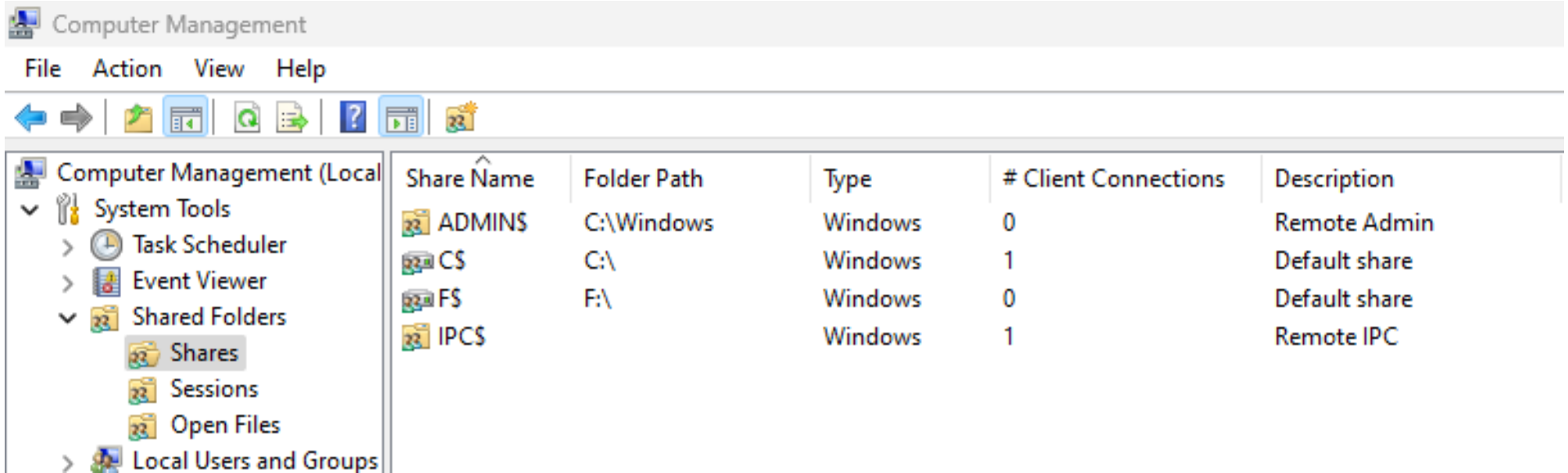
The logo of OpenZFS – open-source implementation of the ZFS filesystem, by OpenZFS, 2016, Wikipedia (https://en.m.wikipedia.org/wiki/File:OpenZFS_logo.svg). Public domain, trademark of OpenZFS.

7. SMB

Server Message Block

Windows använder **SMB** för att dela

- Skrivare
- Filer

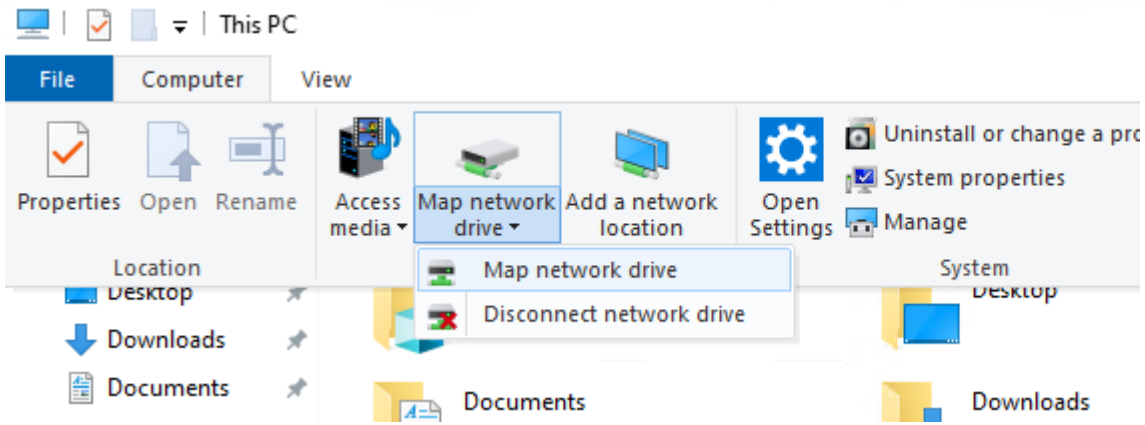


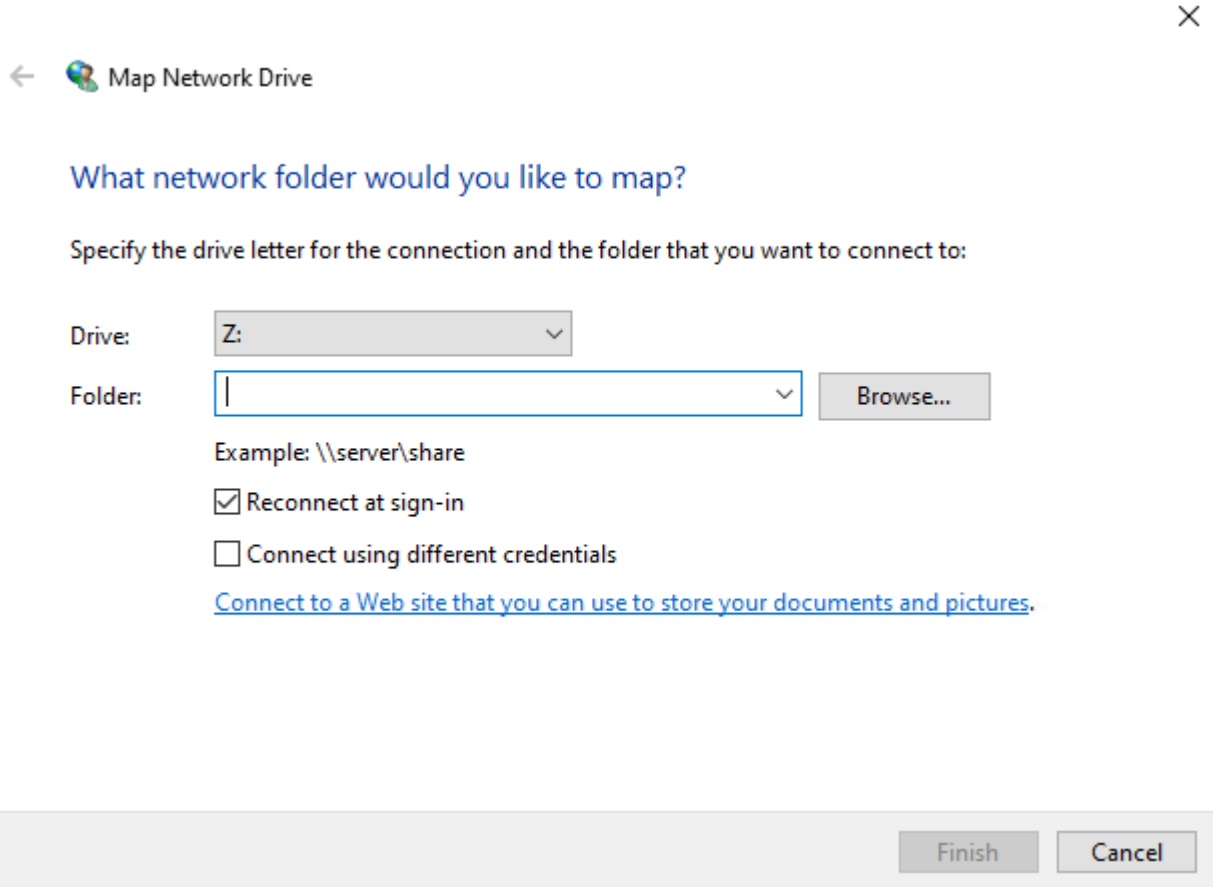
Universal Naming Convention

För att lägga till en file share

\\address\share

\\192.168.1.3\IT



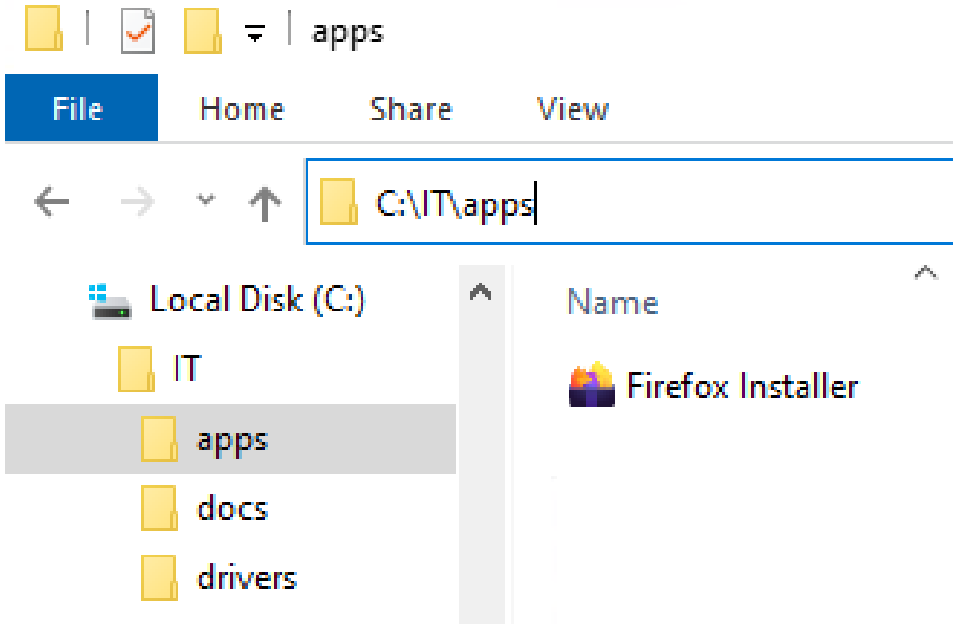


Universal Naming Convention

Ex. Vad är **UNC**-sökvägen till katalogen ”apps”?

- Katalogen ”IT” är delad med namnet ”IT”
- SMB-servern har IP: 192.168.10.7

\\192.168.10.7\IT\apps



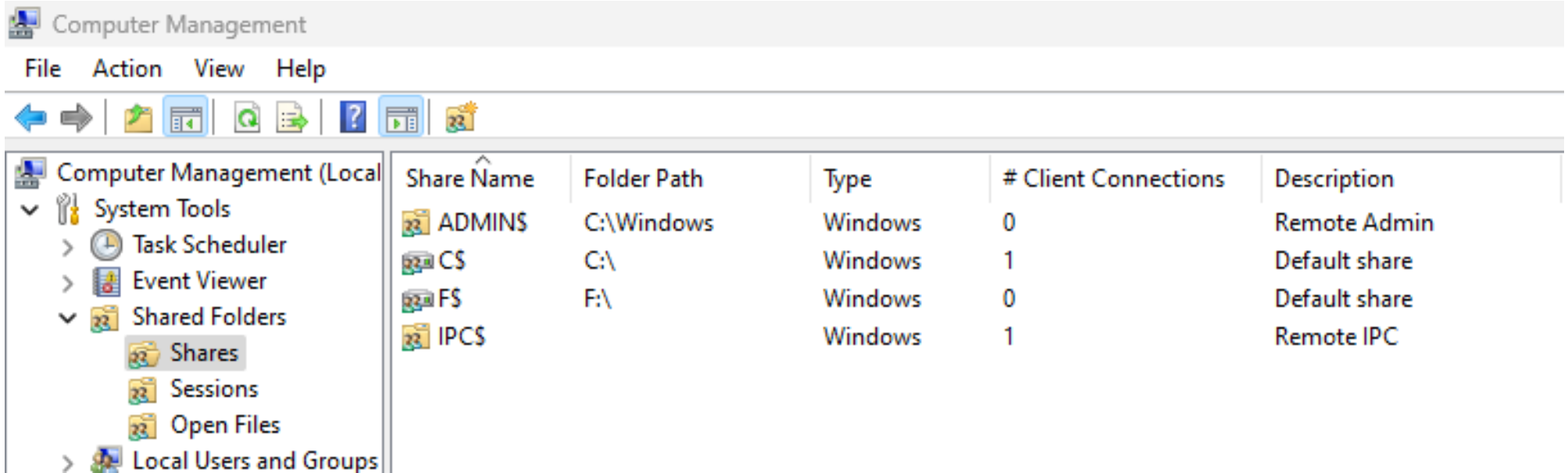
SMB File Shares

Administrativa shares

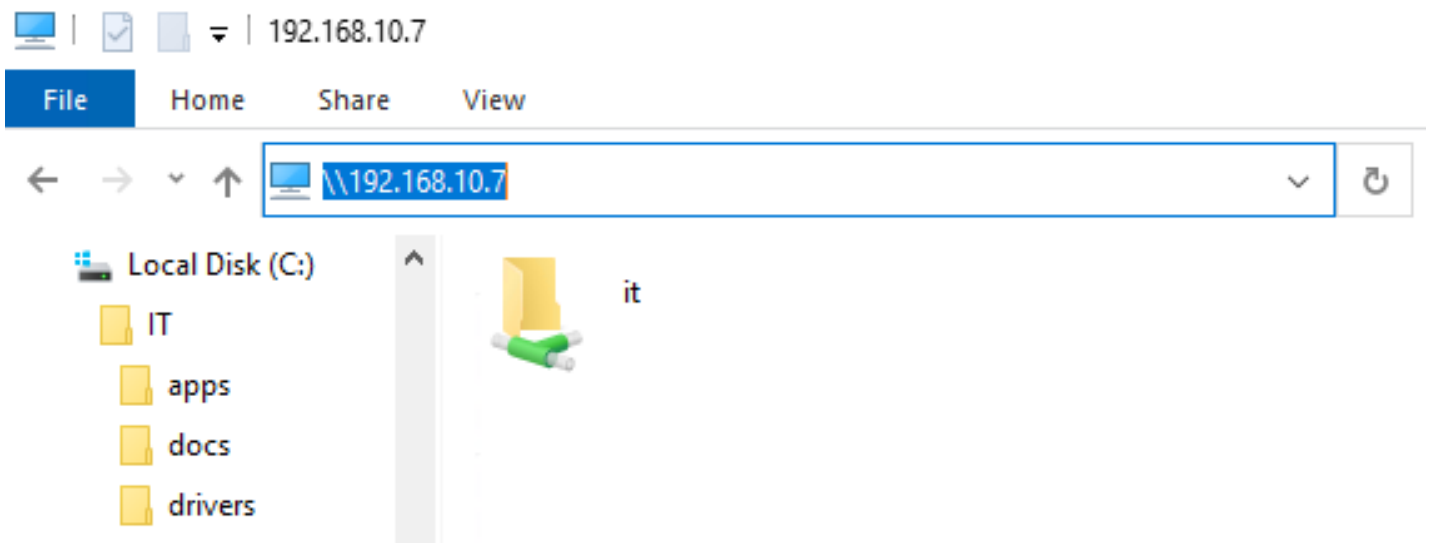
Finns alltid, b.l.a ”C$” (C:)

Hidden shares

Avslutas med ”$”



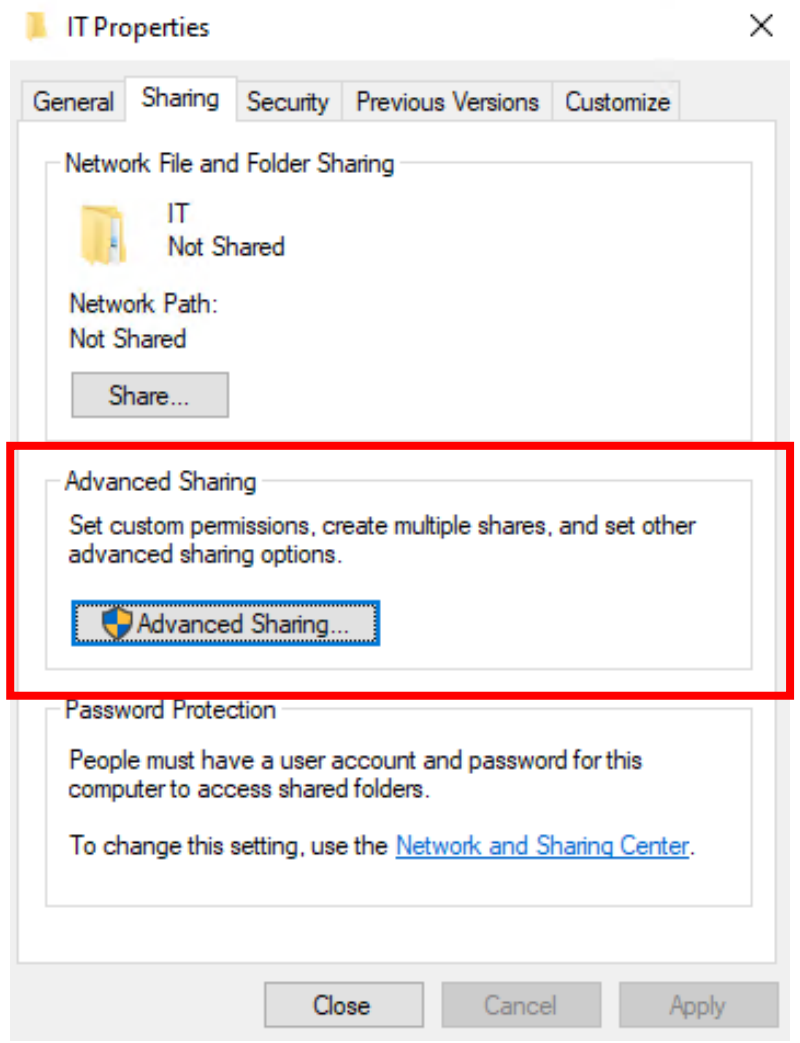
SMB File Shares

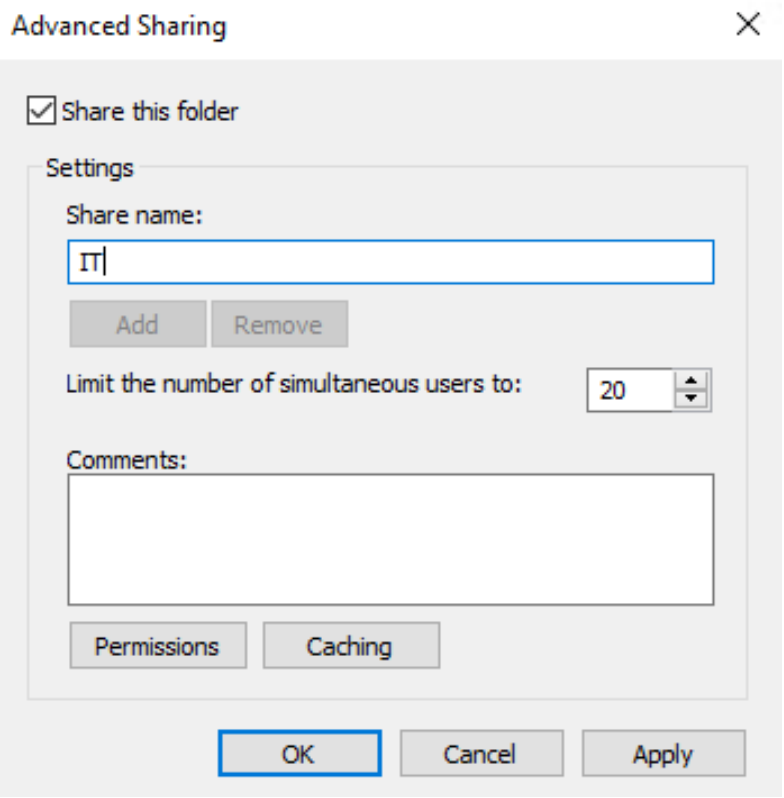


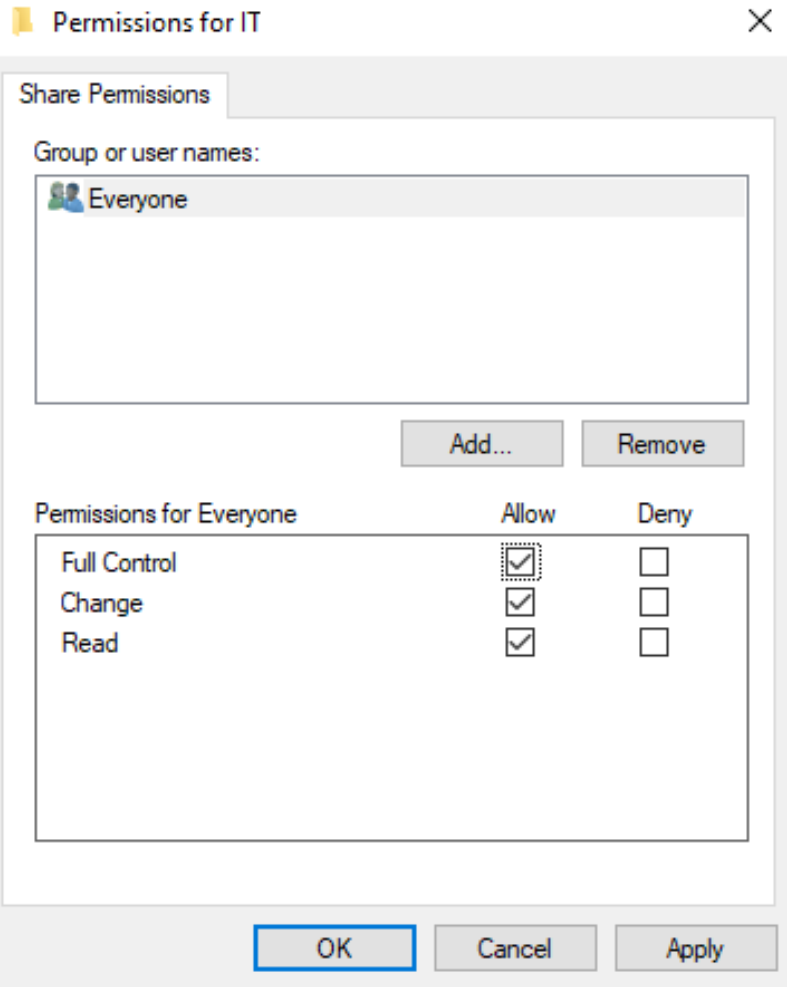
Kan ansluta till en SMB-server och se alla icke-dolda shares

SMB File Shares

Använd **Advanced Sharing**!







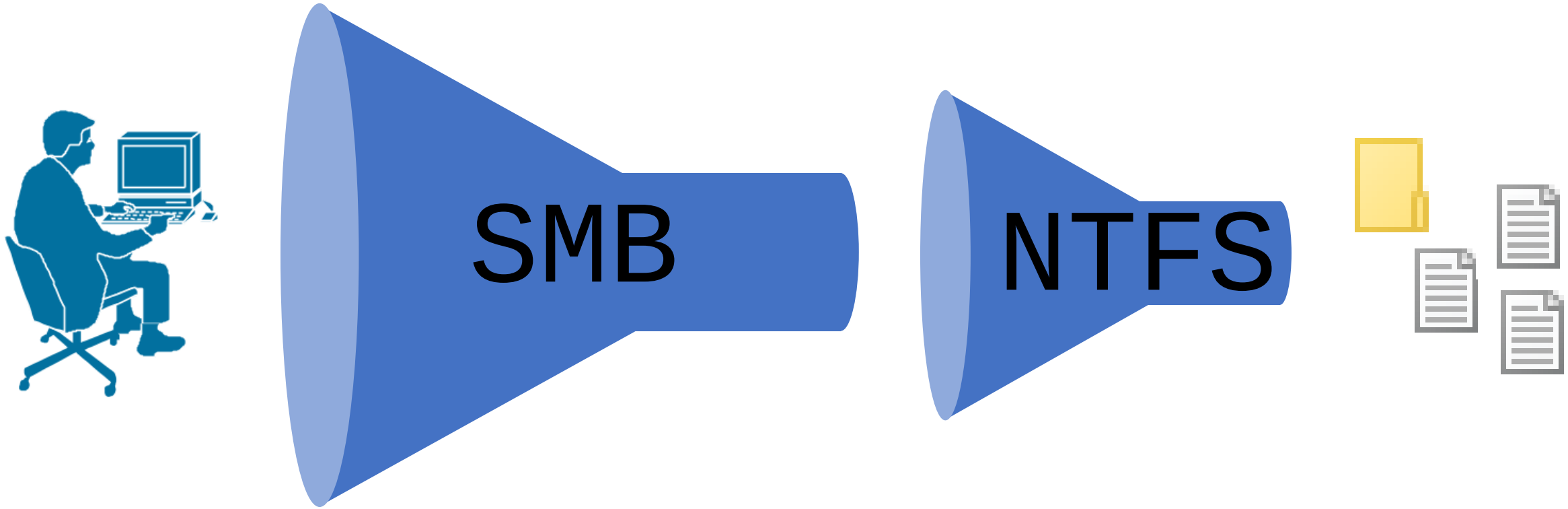
SMB Shares Permissions

Permissions name	Ability granted
Read	See contents of files & folders
Change	**Read**, execute, modify & delete files & folders
Full Control	**Change** + modify permissions

SMB + NTFS

SMB-rättigheter ersätter inte ditt **file shares** filrättigheter…

Måste passera båda!



SMB + NTFS

Ge därför **Everyone Full Control**

Reglera sedan rättigheter med NTFS

Varför?

- Administrativt enklare
- Samma rättigheter oavsett om man når filer lokalt eller över ett nätverk

Till nästa gång

- Läs kurslitteraturen
- Se video(?)
- Labb: fredag

Tack för idag